

HANAN HUSSAIN

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github.com/HananProjects | hanan-portfolio-xi.vercel.app

EDUCATION

University of Saskatchewan | Saskatoon, SK

Bachelor of Science in Computer Engineering | June 2026 | GPA: 3.0/4.0

Relevant Coursework: Software Engineering, Digital Design, Embedded Systems, Verification, Computer Networks

EXPERIENCE

Software Developer — Motion Fitness | Saskatoon, SK | 2026

Python

- Built internal financial dashboards used by the accounting team to review, reconcile, and automate expense work across 5 gym locations, replacing 5 separate spreadsheets with a single consolidated reporting view
- Automated expense and invoice processing end-to-end with Python, cutting per-location processing time from **15 minutes to 5 (3x faster)** and auto-flagging discrepancies before reconciliation, eliminating recurring manual-entry errors
- Designed and shipped an internal HR management system with zero recurring costs that fully replaced a paid third-party HR SaaS subscription (**saving them \$2400 a year**)

Software Developer — Autoholic | Saskatoon, SK | 2026

Next.js 15, TypeScript, SQLite, Railway

- Built and deployed a production invoicing system and business website now in daily use by the shop — client, vehicle, invoice, and expense management
- Designed the relational **SQLite schema** for client, vehicle, invoice, and expense data, with queries handling GST/PST tax calculations and reconciliation
- Owned the full delivery cycle solo: requirements with the owner, design, deployment on Railway, and ongoing support

PROJECTS

FraudGuard – Real Time Fraud Detection Pipeline | 2026

Go · Next.js/TypeScript · Terraform · Azure (Event Hubs, Cosmos DB, Container Apps, SignalR) · GitHub Actions

- Designed and built a distributed event-driven fraud-detection pipeline on **Azure**: A **Go** microservice consumes a live event hubs transaction stream on Azure Container Apps, autoscaling via **KEDA** on Event Hubs lag to zero for fault-tolerant scaling
- Built a rules-based fraud-scoring engine (z-score, velocity, geo checks) — 50% precision, 64% recall on labeled test data
- Delivered real-time, consistent dashboard updates via **Cosmos DB** with a NextAuth-gated Next.js/TypeScript frontend

AI Code Review Pipeline | 2026

Python, CrewAI, Claude API, FastAPI, GitHub API, GCP Cloud Run, Docker

- Built a multi-agent AI code review system using CrewAI and Claude API that automatically reviews GitHub PRs via webhooks; deployed to **GCP Cloud Run** with Docker — serverless, scales to zero, permanent public URL
- Engineered and orchestrated **3 parallel AI agents** (security, performance, style) running concurrently via ThreadPoolExecutor, reducing review time by **3x vs sequential execution**
- Integrated GitHub Review API to post **inline comments pinned to specific lines of code**

Two-Way Portable English-to-ASL Translator (Capstone) | Sept 2025 - Apr 2026

Raspberry Pi 5, Python, MediaPipe, VOSK STT, C, Linux, I2S/I2C/CSI, Edge Inference

- Designed and integrated a real-time bidirectional ASL-English translation system on Raspberry Pi 5; achieved **end-to-end latency under 2 seconds** with fully offline operation across **100+ word vocabulary**
- Developed a key-point-based gesture recognition pipeline using Python and MediaPipe; achieved **>85% classification accuracy** across gesture classes through landmark extraction, inference logic, and token mapping
- Architected a modular, fault-tolerant multi-threaded system managing **5 concurrent subsystems** (camera, microphone, display, speaker, control) on a single embedded platform within a **\$500 hardware budget**
- Led system-level **verification** on a 4-person team using **sprint-based** iterative development: timing analysis, latency benchmarking, and integration debugging
- Built a pytest suite covering gesture-classification and token-mapping logic to validate accuracy and catch regressions across development iterations

TECHNICAL SKILLS

Languages: C, C++, Python, Java, Verilog, TypeScript/JavaScript (React, Node.js), HTML/CSS, SQL

Frameworks & Tools: FastAPI, Django, Next.js, Tailwind CSS, Linux, Bash, Git, Docker, GCP Cloud Run, Railway, Azure, REST APIs, Pytest, Agile/Scrum, CI/CD (Github Actions)

CERTIFICATIONS

Google Cloud Fundamentals - Core Infrastructure (2025) | GitHub Foundations Certificate (2024) | AWS Certified Cloud Practitioner (2026) | AWS Certified AI Practitioner (2026) | Microsoft Azure Fundamentals (AZ-900)